a Program composition under authenticated actor training exposure

Recorded runs; unequal work. Program occurrences, distinct signatures and reactions are different units.

Policy	Seen contexts	Unseen contexts	Unseen signatures	Reactions with unseen programs
MCTS	5,192	6,288	3,890	4,952
Greedy I	863	478	249	467
Greedy II	866	498	253	475

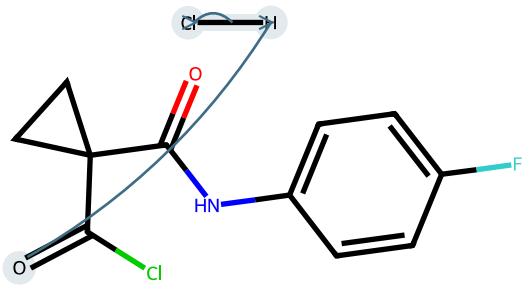
b Whole-reaction precedent for seen versus unseen MCTS programs

Composition	Distinct reactions	A/B-supported reactions	Supported targets
Seen	3,195	28	24
Training-unseen	4,952	11	11

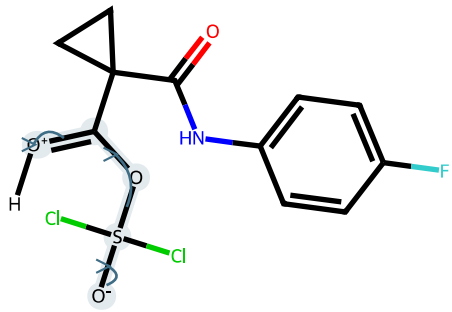
A reaction may have both seen and unseen programs; reaction counts are not additive. Reference C0/C1/C2: Unknown.

c Selected recorded inverse FLOW actions, not inferred physical intermediates

Action 2 · 2 recorded pair moves



Action 7 · 3 recorded pair moves



Selected complete program: unseen ordinary. All actions retained in linked plates.

Source-backed net chemistry does not establish the exact microscopic electron-arrow order.

Training-unseen electronic composition is not new chemistry, strict-H2 generalization or a reference C1 label.